

SPMS

Smart Power Monitoring Solution Project

Introduction

This project aims to optimize the fault detection and maintenance process of MSAN (Multi-Service Access Node) systems by introducing an intelligent IoT-based monitoring solution. At present, every MSAN fault requires a field team visit to identify the root cause, even in cases of simple power loss or voltage fluctuations. These unnecessary site visits result in significant operational costs and resource inefficiencies.



Problem

- MSAN faults happens due to following reasons -
 - CEB Power loss
 - Breaker OFF from power distribution board
 - MSAN Fault
- Currently, a physical team need to visit to the fault MSAN location to identified the fault type or reason. However, in the event of CEB power failure, Power fluctuation there is no need to visit the Faulty location.
- The Cost for these unnecessary movements is a considerable amount compared to operation cost.

Sample time period = 4/20/2024 to 5/20/2024

No. of MSANs in KY RTOM area = 335

No. of visits = 173

Visit rate = 52%

Visits for 20 units for 6 months = 61.97014925 = 62

Average distance travelled per visit (both ways in km) = 48

Average time consumed per visit (hrs) = 2

As per Job Costing Unit, value of 1 man Hr (Rs.) = 922

Fuel consumption for SLT Double Cabs = 8 km / L

Per Liter cost of Auto Diesel (Rs.) = 333

Total cost saving for POC for 6 months

= Staff cost + Vehicle Cost (Excluding Repair & Maintenance) = 228656 + 123876

= LKR 352,532.00

Solution

1. Develop IOT device to detect presence of 230VAC.
2. Detect voltage fluctuations within given range.
3. Implement control signals for breakers.
4. Make a dashboard to visualize voltage status of each device.
5. Generate alarms based on the events (voltage loss, fluctuations and device offline)
6. Implement external API to integrate with current MSAN Manage Dashboard.

Pilot Test - 500 Data Points

Location	Mattakkuliya MSAN
Time span	5 days + 0:36:44
First timestamp	10/15/2025 4:28:23 PM
Last timestamp	10/20/2025 5:05:07 PM
Expected data points	502
Observed data points	502
Availability (%)	100%

Interval Accuracy

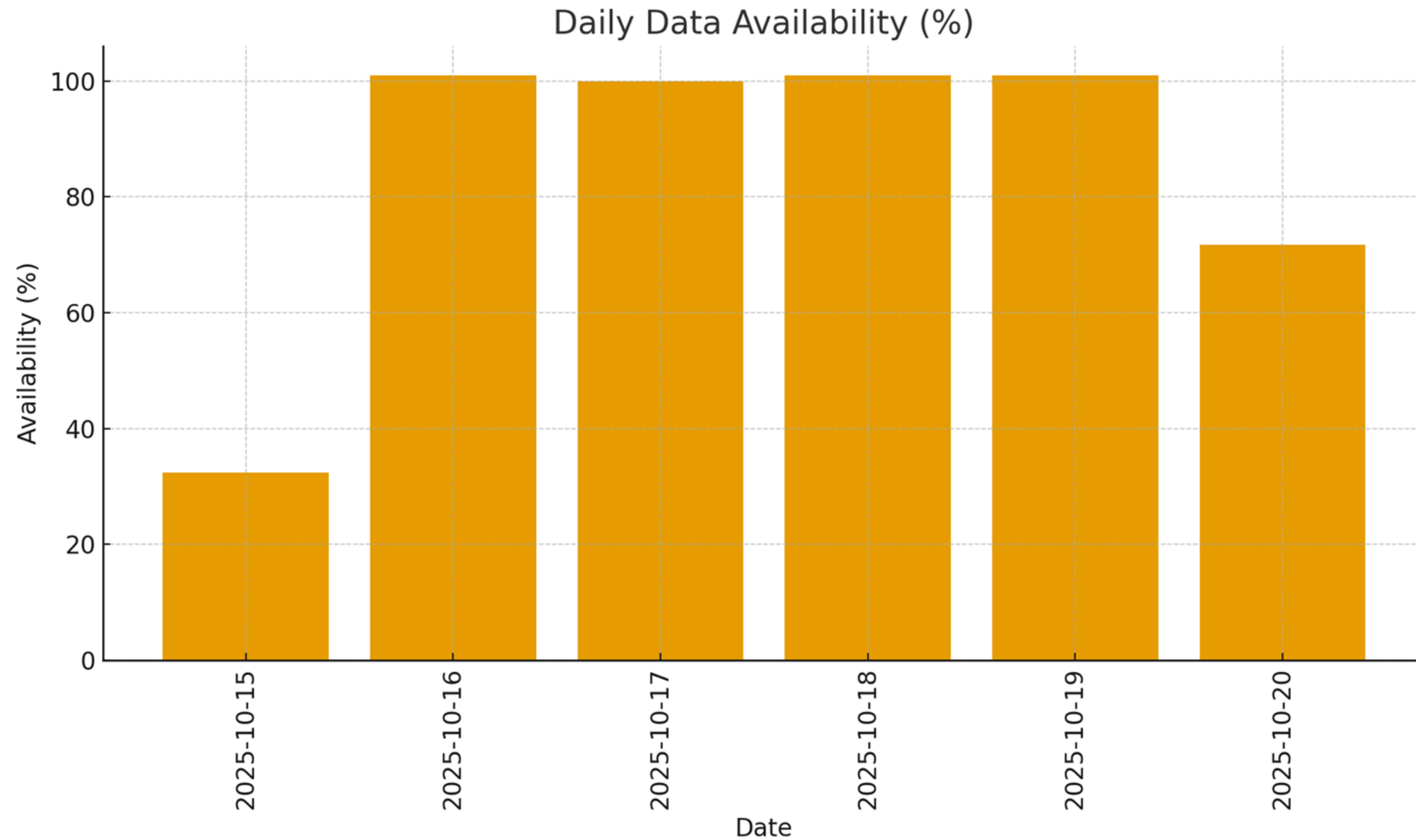
Ideal interval = 14min & 30sec (870sec)

Average interval (s)	866.67
Median interval (s)	870
MAE vs ideal (s)	4.14
Within ± 5 s of ideal (%)	97.60%
Within ± 30 s of ideal (%)	99.40%

Availability

Date	expected approx	observed	availability %
10/15/2025	99	32	32.32%
10/16/2025	99	100	101.00%
10/17/2025	99	99	100.00%
10/18/2025	99	100	101.00%
10/19/2025	99	100	101.00%
10/20/2025	99	71	71.72%
Expected points		502	
Observed points		502	
Availability (%)		100.00%	

Availability



Voltage vs Timestamp

